

Vistula University

The Faculty of Computer Engineering, Graphic Design and Architecture

Program of study Computer Science

Muhammed Fariz Palli Valappil

Student number 74944

***THE DISPROPORTIONATE IMPACT
OF POST-TRAINING QUANTIZATION
(PTQ) ON MULTILINGUAL
REASONING: A COMPARATIVE
ANALYSIS OF ENGLISH VS. POLISH
DEGRADATION IN LLAMA-3***

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dr hab. Paweł Gburzyński, Profesor Uczelni

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Introduction

Large language models (LLMs) like the Llama-3 family are incredibly powerful, but their sheer size makes them practically impossible for most people to run locally on standard consumer hardware. Post-Training Quantization (PTQ) solves this bottleneck by compressing the model's weights from standard 16-bit floating-point precision down to 8 or 4 bits, drastically shrinking the memory footprint required for inference. However, this mathematical compression is not free. Forcing massive neural weights into lower bit-depths inherently destroys information. While the AI engineering community actively measures the general degradation caused by PTQ, there remains a critical lack of understanding regarding exactly which specific linguistic capabilities are sacrificed in the process.

Currently, the vast majority of research on model compression evaluates performance almost exclusively using standard English benchmarks. Testing a multilingual model only in English is a bit like testing a car only on dry roads and calling it road-safe. The conditions where failure is most likely are exactly the ones being skipped. Because modern LLMs are pre-trained on internet-scale data that heavily skews toward English, their ability to process and reason in other languages relies heavily on a mechanism known as cross-lingual transfer. While these models can demonstrate impressive reasoning capabilities in languages like Polish at full precision, it is highly probable that these capabilities are structurally weaker beneath the surface. If a language represents a lower priority in the initial training distribution, the neural pathways representing it may be significantly more fragile when the model undergoes destructive weight compression.

This thesis investigates that exact vulnerability. The core hypothesis is that under aggressive 4-bit quantization, Llama-3's reasoning accuracy in Polish will degrade at a faster and more severe rate than its accuracy in English. To explore this dynamic, this research introduces the concept of Secondary Competency, a language capability developed primarily through cross-lingual transfer rather than primary training data. A Secondary Competency can look perfectly healthy at full precision. The problem is that compression does not affect all capabilities equally. The sparser ones which are the ones built on borrowed cross-lingual scaffolding rather than direct training, are the first to crack when the model gets squeezed.

Declaration of AI Usage: In the preparation of this thesis, AI-based tools, specifically Google Gemini, and Claude by Anthropic were used for the purposes of debugging Python code during the experimental phase and performing standard grammar checks on the manuscript. All

experimental design, benchmark selection, data collection, statistical analysis, and the written content presented herein are the original work of the author.

The structure of this thesis follows the logical progression of the experiment. Chapter 1 establishes the theoretical background required to understand the problem, explaining Transformer architecture, the mechanics of cross-lingual transfer, and how PTQ methods physically compress data. Chapter 2 details the methodology, specifically highlighting an unexpected finding regarding the systemic exclusion of Polish from major evaluation benchmarks, which necessitated the selection of the Belebele dataset. Chapter 3 breaks down the experimental results, introducing the novel "Convergence Effect" to explain how 4-bit quantization effectively erases secondary language performance advantages, followed by a discussion of the real-world implications for developers and end-users. Finally, the Conclusion summarizes the empirical findings and outlines necessary directions for future research.

The theoretical framework of this paper draws on a literature base spanning from the foundational Transformer architecture (Vaswani et al., 2017) through to the most recent developments in multilingual quantization research published in early 2026. The specific intersection of quantization and multilingual performance is a remarkably recent area of academic focus. The first comprehensive, systematic study of quantization's impact across languages was published only in mid-2024 (Marchisio et al., 2024), meaning this thesis is positioned at the direct frontier of the field. A January 2026 study by Chimoto et al. directly corroborates the systemic calibration failure this thesis investigates, while a contrasting result from Borgersen & Goodwin (2025) who found no significant multilingual degradation for Norwegian under GGUF k-quantization, underscores the need for language-specific and method-specific analysis that this research provides. Primary sources are drawn from major machine learning conferences (ACL, NeurIPS, EMNLP) and pre-print archives (arXiv) to ensure the absolute current state of both quantization techniques and multilingual evaluation methodologies is accurately represented.

Chapter 1. Theoretical Background

To understand how post-training model compression disproportionately affects secondary languages, it is first necessary to establish how these models process language at a fundamental, architectural level. This chapter reviews the structural foundations of Large Language Models (LLMs), the mathematical mechanics of cross-lingual transfer, and the engineering principles behind Post-Training Quantization (PTQ).

1.1 Large Language Models

The current generation of artificial intelligence is fundamentally defined by the Transformer architecture, a neural network design introduced by Vaswani et al. (2017). Prior to the advent of Transformers, natural language processing (NLP) relied heavily on sequential models such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks. These legacy architectures processed text linearly, word by word, which created a severe computational bottleneck for training on large datasets. On top of that, sequential processing caused models to suffer from vanishing gradients, effectively making them "forget" earlier parts of a long text sequence.

Transformers solved this inherent limitation through a breakthrough mechanism called "Self-Attention." Instead of reading a sentence from left to right, a Transformer processes all tokens in a sequence simultaneously. The self-attention mechanism calculates a mathematical weight matrix that determines how strongly every word in a sentence relates to every other word, regardless of their physical distance in the text.

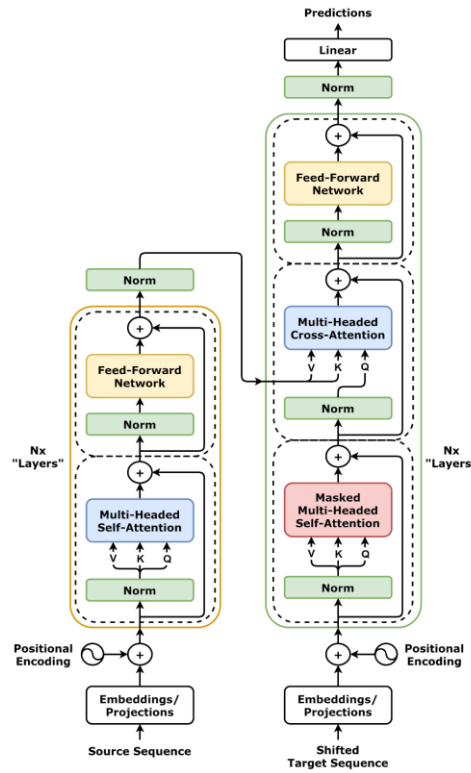


Figure 1.1. The standard Transformer architecture, highlighting the self-attention mechanism.

Source: Vaswani et al. (2017).

For instance, in the phrase “*The bank of the river,*” the self-attention mechanism assigns a strong contextual association between the token *bank* and *river*, guiding the model to interpret *bank* as a geographic feature rather than a financial institution. This capacity to preserve long-range context without relying on sequential processing has enabled researchers to parallelize training across thousands of GPUs, enabling to increase both model size and the volume of training data substantially.

Modern LLMs operate at their core as massive, highly complex probability engines. During their pre-training phase, they have been fed terabytes of text scraped from the internet, digitalized books, and code repositories. Their primary training objective is simple, which is next-token prediction (Brown et al., 2020). By repeating this task of prediction billions of times across vast datasets and continuously adjusting their internal weights through backpropagation, the model starts to learn the grammar, factual relationships, reasoning patterns, and structural biases present in their training dataset.

For this specific research, the Llama-3 family of models, specifically Meta-Llama-3-8B-Instruct was selected. Released by Meta AI (2024), Llama-3 represents the current state-of-the-art for open-weight foundational models. The 8B (8 billion parameter) variant is particularly relevant for this experimental setup. It is small enough to be manipulated, quantized, and evaluated on a single consumer-grade GPU (such as the 15GB VRAM Tesla T4 utilized in this study), yet architecturally sophisticated enough to demonstrate complex multilingual reasoning. Furthermore, the "Instruct" variant has been heavily fine-tuned using Reinforcement Learning from Human Feedback (RLHF) to follow user commands strictly and output formatted responses, making it highly reliable for automated benchmark evaluation scripts (Touvron et al., 2023).

The Latent Space and the Residual Stream

In order to understand why quantization harms some linguistic capabilities more than others, we must examine how information is actually encoded and propagated inside a Transformer.

When the text is tokenized, each token is immediately mapped to a high-dimensional vector of learned numbers. Each token in the Llama-3-8B architecture is represented by a vector of **4,096 floating-point numbers**. These 4,096 numbers do not have a direct human-readable meaning. They are coordinates in an abstract, 4,096-dimensional mathematical space that the model learned to build during its pre-training. This is called latent space, a vast geometric space in which the mathematical distance between two vectors indicates their semantic connection in meaning.

The latent space is not uniformly filled. In some regions the geometry is tight and densely packed, which indicates to concepts the model has seen millions of times English grammar, common English idioms, standard English sentence structures. When we look into Polish, the same concepts exist in that space too, but the surrounding geometry is looser, which indicates far fewer training examples. For example, an English phrase like "the bank approved the loan" is surrounded by thousands of geometrically close neighbors, shaped by millions of similar training examples. An equivalent Polish phrase exists in the same latent space, but its local geometry is defined by far fewer examples during the training, which makes the surrounding region distant and more loosely structured.

Information does not simply sit in the latent space statically, it flows dynamically through the model's 32 processing layers via what researchers call the residual stream. The

residual stream is a single 4,096-dimensional vector that begins as the token embedding and is progressively enriched and improved as it passes through each Transformer layer. At every layer, two operations write their output back into this shared stream. First, the multi-head self-attention operation, which includes contextual information from all other tokens in the sequence. Second, the feed-forward network, which applies a learned non-linear transformation. By the time the residual stream exits the final layer, it encodes the full contextual representation of the token and this is the final vector that the model feeds into a probability distribution to generate or predict the next output.

Here is where quantization matters. Every operation inside the model involves multiplying the residual stream by weight matrices. Those matrices carry rounding errors under 4-bit compression. Individually small errors. But they apply at every layer for all the 32 layers, and for a Polish prompt that generates 45% more tokens than the English equivalent, those small errors gets applied far more often. Under 4-bit NF4 quantization, every element of these weight matrices is stored only one in 16 possible values. Every matrix multiplication therefore introduces a small rounding error. Across 32 layers of residual stream updates, these errors accumulate. The important question and the core empirical finding of this thesis is whether these accumulated errors damage sparse regions of the latent space (Secondary language capacity like Polish) more severely than dense regions (primary language capacity like English). The experimental results presented in Chapter 3 confirm that, at the more aggressive 4-bit compression level, the models tend to do that.

1.2 Multilingual Competency in LLMs

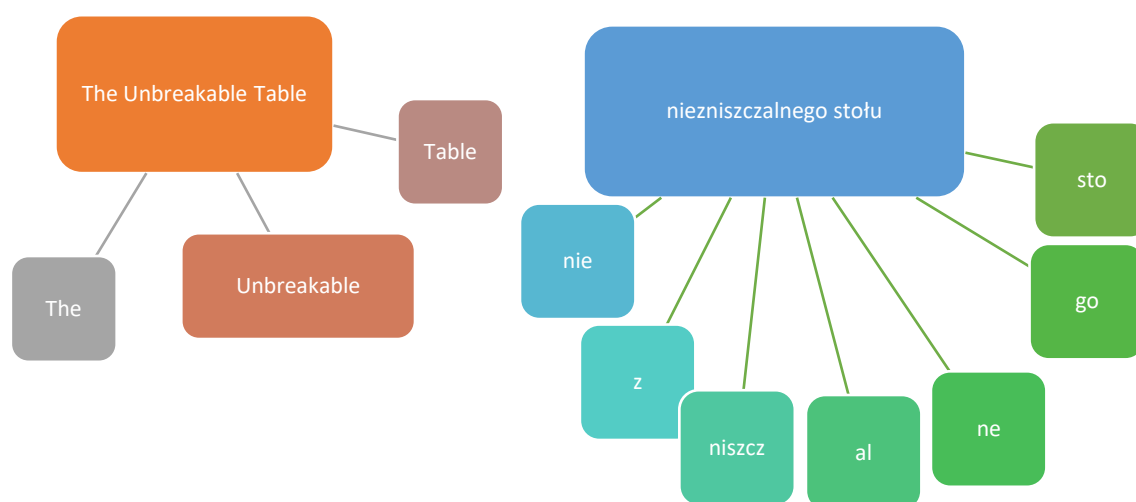
A fascinating and highly relevant property of foundational LLMs is that they are not always explicitly programmed with the grammatical rules or vocabularies of different languages. Instead, multilingual competency is a property that develops naturally from the pre-training data distribution. Because the internet is inherently multilingual, the massive Common Crawl datasets used to train models like Llama-3 undoubtedly contain billions of tokens of text in French, Spanish, Chinese, Polish, and other languages.

However, this data distribution is massively skewed. High-resource languages, specifically English, account for the overwhelming majority of LLM training data. Despite this severe imbalance, models consistently demonstrate a remarkable ability to process prompts and reason in languages that make up only a tiny fraction of their training set. This capability is driven by a phenomenon known as cross-lingual transfer (Conneau et al., 2020).

During the pre-training phase, the neural network maps concepts into a high-dimensional mathematical vector space (the latent space). Because different human languages are simply different encoding systems describing the exact same physical reality, the network eventually learns to align the underlying mathematical representations of words. The English word "table" and the Polish word "stół" are eventually mapped to the exact same abstract concept vector in the model's latent space. Therefore, if the model learns complex logical reasoning pathways primarily through processing English text, it can transfer that reasoning capability to Polish queries by routing the Polish input through its shared, language-agnostic conceptual framework (Shi et al., 2022).

Despite the efficiency of cross-lingual transfer, a critical architectural friction point exists at the very first layer of the model: tokenization. LLMs do not read text as whole words; they break raw text strings down into smaller sub-word units called "tokens" using algorithms like Byte Pair Encoding (BPE). Because the tokenizer's vocabulary is statistically optimized to compress the dominant training language (English), common English words are usually mapped to highly efficient single tokens.

Figure 1.2. Illustration of the tokenization penalty: English words mapping to single tokens versus morphologically rich Polish words fracturing into multiple sub-word tokens.



Source: Own elaboration.

The tokenization problem with Polish is concrete, not theoretical. It can be measured directly. To illustrate this problem, the following table presents parallel English and Polish sentences evaluated through the Llama-3-8B-Instruct tokenizer, measuring the token count required to encode the same semantic content in each language.

Table 1.1. Tokenization penalty: parallel English and Polish sentences in the Llama-3 BPE tokenizer.

English Sentence	EN Tokens	Polish Sentence	PL Tokens	Ratio
The student reads a book about history.	9	Student czyta książkę o historii.	13	1.44x
She walked quickly through the crowded market.	9	Szła szybko przez zatłoczony rynek.	14	1.56x
The results confirm the research hypothesis.	8	Wyniki potwierdzają hipotezę badawczą.	13	1.63x

Source: Own elaboration based on Llama-3-8B-Instruct tokenizer output.

The data reveals a consistent tokenization overhead of approximately 1.4 to 1.6 times more tokens required to encode equivalent Polish content in English. This overhead is not merely a storage inconvenience; it has direct and compounding mathematical consequences for the self-attention mechanism.

Self-attention has a quadratic relationship with sequence length, $O(n^2)$. Double the tokens and you get four times the computations, not two. So, when Polish requires 1.5 times more tokens for the same content, the attention mechanism runs approximately 2.25 times as many calculations. Under 4-bit quantization, each of those calculations carries rounding error. The longer Polish sequences do not just require more work, they accumulate more error per prompt, structurally. For a Polish reading comprehension passage requiring, for example, 150 tokens where an English passage requires 100, the self-attention calculation involves 22,500

pairwise token relationships versus 10,000 for English, a 125% increase in the number of individual precision-sensitive calculations performed per prompt.

Under full FP16 precision, the attention mechanism is mathematically robust enough to handle this extended sequence without significant quality degradation. The self-attention weights are computed to 16-bit precision across all token pairs, and the model's residual stream accurately accumulates the contextual signal across the full sequence length. However, under 4-bit NF4 quantization, each individual attention calculation carries a small rounding error. Across 22,500 Polish token-pair calculations versus 10,000 English ones, the cumulative rounding error introduced into the residual stream is not merely additive, it compounds across these calculations, structurally degrading the model's ability to maintain coherent contextual reasoning over the extended Polish sequence. This compounding quantization-tokenization interaction represents the second key mechanism, alongside weight sparsity, driving the Convergence Effect documented in Chapter 3.

To frame the core hypothesis of this thesis, it is necessary to formalize the concept of Secondary Competency. A primary competency (such as English in the Llama-3 architecture) is deeply entrenched. Its neural pathways are densely reinforced by massive volumes of high-quality training data, and its input is processed through highly efficient, single-word tokenization.

Conversely, a Secondary Competency (such as Polish) relies heavily on the bridge of cross-lingual transfer. It is a fundamentally derived capability. While an LLM might successfully answer a complex logical question in Polish at full FP16 precision, the neural weights supporting this reasoning are mathematically sparser and less robustly reinforced than their English equivalents. The central argument of this research is that when a model's overall representational capacity is violently reduced through quantization, these sparser, secondary representations will be structurally unable to survive the compression, resulting in disproportionate capability degradation.

1.3 Post-Training Quantization

The primary barrier to deploying powerful LLMs is their immense memory requirement. A model's weights are typically stored in 16-bit floating-point precision (FP16). For an 8-billion parameter model like Llama-3-8B, simply loading the model into a GPU requires approximately 16 Gigabytes of Video RAM (VRAM) before any inference calculations even begin. This effectively locks advanced AI out of standard consumer hardware.

Quantization is a mathematical compression technique designed to solve this bottleneck. It works by mapping continuous, high-precision floating-point numbers into a smaller set of discrete, lower-precision values (such as 8-bit or 4-bit integers). While Quantization-Aware Training (QAT) incorporates this compression during the initial model training phase, Post-Training Quantization (PTQ) is applied to pre-trained, full-precision models. PTQ is highly favored in the engineering community because it does not require the massive computational resources necessary to retrain the model from scratch (Nagel et al., 2020).

However, reducing bit-depth inherently causes information loss, which degrades model performance. Standard quantization often fails in LLMs because of "outlier features", specific neural weights that are significantly larger than the rest of the distribution. If these outliers are aggressively compressed, the model's fundamental logic collapses. To counter this, advanced PTQ methods have been developed and integrated into standard deployment pipelines via libraries such as HuggingFace's bitsandbytes.

For the 8-bit condition (Q8) evaluated in this thesis, the LLM.int8() method was utilized. This algorithm isolates the mathematical outliers and processes them in full FP16 precision, while quantizing the vast majority of the standard weights into 8-bit integers (Dettmers et al., 2022). This allows for a near 50% reduction in memory with historically minimal performance degradation on English benchmarks.

For the aggressive 4-bit condition (Q4), the NormalFloat 4-bit (NF4) data type was utilized. Introduced alongside the QLoRA framework, NF4 is an information-theoretically optimal data type specifically engineered for weights that follow a normal distribution, which is characteristic of trained neural networks (Dettmers et al., 2023).

To ensure complete reproducibility of these specific quantization conditions, the HuggingFace BitsAndBytesConfig class was utilized in the experimental setup.

The Mathematics of Information Loss

To fully appreciate why quantization degrades model performance, it is necessary to understand the mathematical operation at the heart of the process. In standard FP16 storage, a single neural weight value can represent any of 65,536 distinct numerical values across a continuous range. When that same weight is compressed to 4-bit NF4, it must be mapped to one of only **16 possible discrete values**. The quantization process selects the nearest representable

value and discards the difference. This discarded difference is the *quantization error*, and it is introduced silently into every single one of the model's billions of weight parameters.

For the Llama-3-8B model, this means that approximately 8 billion individual weight values, each originally stored at 16-bit precision, are each individually rounded to their nearest 4-bit equivalent. The statistical distribution of these individual rounding errors tends to be small and random for weights that follow the expected normal distribution, which is precisely why NF4 was designed for normally distributed weights. However, weights that deviate from this distribution, or weights that encode fine-grained distinctions between sparse concepts (such as Polish morphological patterns), will experience systematically larger quantization errors.

The standard metric used in the quantization research community to measure aggregate precision loss is **perplexity**, this is a measure of how surprised the model is by a text sequence it is asked to predict. Lower perplexity indicates higher confidence and better language modeling quality. Chimoto et al. (2026) use perplexity as their primary metric, demonstrating that English-only calibration sets produce higher (worse) non-English perplexity compared to multilingual calibration. The present study measures the downstream consequence of this perplexity increase directly: reasoning accuracy on a structured comprehension task, which provides a more operationally meaningful metric for real-world deployment quality than raw perplexity.

It is also worth noting the existence of alternative PTQ methods not evaluated in this thesis. GPTQ (Frantar et al., 2022) and AWQ (Lin et al., 2024) are the two most widely deployed alternatives to bitsandbytes NF4, used extensively in local inference frameworks such as llama.cpp and vLLM. Both methods incorporate an optimization step that minimizes the quantization error with respect to a small calibration dataset, which is something that bitsandbytes NF4 does not perform in the same way. This architectural distinction is central to the calibration mismatch hypothesis advanced by Chimoto et al. (2026). Because GPTQ and AWQ optimize against calibration data, the language of that data directly shapes which weights are preserved with higher fidelity. The bitsandbytes NF4 method evaluated in this thesis applies a more uniform compression strategy, making its behavior on secondary languages an important independent data point in the broader multilingual quantization research landscape.

```
# Q8 Implementation
bnb_config_8bit = BitsAndBytesConfig( load_in_8bit=True )

# Q4 Implementation (NF4)
bnb_config_4bit = BitsAndBytesConfig(
    load_in_4bit=True,
```

```

        bnb_4bit_quant_type="nf4",
        bnb_4bit_compute_dtype=torch.float16
    )

```

Table 1.2 outlines the three specific quantization conditions evaluated in this research, detailing their precision levels, underlying algorithms, and the approximate VRAM required to load the Llama-3-8B-Instruct model on the Tesla T4 GPU used for testing.

Table 1.2. Quantization conditions evaluated in the experimental setup.

Condition	Precision	Algorithm	Approx. VRAM Footprint
Baseline	FP16 (16-bit)	Standard Float	~16.0 GB
Q8	INT8 (8-bit)	LLM.int8()	~8.5 GB
Q4	NF4 (4-bit)	NormalFloat 4-bit	~5.5 GB

Source: Own elaboration based on bitsandbytes implementation metrics.

1.4 Polish as a Secondary Language Test Case

The Polish language serves as an optimal test case for examining the fragility of Secondary Competencies under quantization. Demographically, Polish is a major European language with approximately 45 million native speakers and primarily concentrated in Central Europe. It is an official language of the European Union, meaning its automated processing holds significant legal, commercial, and social relevance within the broader tech ecosystem.

Linguistically, Polish is a very difficult, morphologically complex West Slavic language. Unlike English, which relies on strict word order and prepositional phrases to convey meaning, Polish utilizes a system of seven grammatical cases (nominative, genitive, dative, accusative, instrumental, locative, and vocative). The spelling of a core noun changes dynamically depending on its role in a sentence. For example, the English noun "table" remains static regardless of context. In Polish, the noun "stół" yields multiple inflected forms: *stół* (of the table), *stółowi* (to the table), *stółem* (with the table), and *stole* (about the table). Beyond that, the language features complex verbal aspects, grammatical genders, and a relatively free word order.

For Natural Language Processing, this morphological richness is fundamentally hostile to standard tokenization models optimized for English. A Polish sentence mathematically requires more tokens to encode the exact same semantic payload as an English sentence. When

a model operates at full FP16 precision, the self-attention mechanism is generally powerful enough to track these extended, fragmented token sequences and successfully deduce the logic. However, under aggressive 4-bit quantization, the mathematical precision of the attention mechanism is severely degraded. This thesis posits that the compounding error of reduced precision multiplied over longer, fragmented token sequences make Polish inherently more vulnerable to PTQ than English.

Polish has 45 million native speakers and is an official EU language. It has an active domestic NLP research community, its own benchmark suite, and a significant digital presence. None of that has translated into consistent inclusion in the major multilingual evaluation datasets used to test foundational models globally. That gap between demographic significance and evaluation representation is precisely what this thesis is designed to document. While dedicated domestic benchmarks exist, such as the KLEJ benchmark (Rybak et al., 2020) and various PolEval tasks, these are rarely utilized by international AI developers to test foundational models. As a result, the global AI community routinely releases heavily compressed, quantized models into the European market without empirical proof of their structural stability in languages like Polish.

Polish Morphological Complexity: A Quantitative Illustration

To make the tokenization penalty concrete rather than abstract, we need to examine the full morphological paradigm of a single Polish noun. Table 1.3 presents the complete grammatical inflection of the Polish word *stół* (table) across all seven grammatical cases in both singular and plural forms. Each of these forms represents a distinct surface string that the Llama-3 tokenizer must process as a unique or partially unique token sequence.

Table 1.3. Complete declension paradigm of the Polish noun "stół" (table) across all grammatical cases.

Case	Singular	English Equivalent	Plural	English Equivalent
Nominative	stół	table (subject)	stoły	tables
Genitive	stołu	of the table	stołów	of the tables
Dative	stolowi	to/for the table	stolom	to/for the tables
Accusative	stół	table (object)	stoły	tables
Instrumental	stółem	with/by the table	stolami	with the tables
Locative	stole	about/on the table	stolach	about the tables
Vocative	stole	O table!	stoły	O tables!

Source: Own elaboration based on Polish grammar reference (PWN, 2020).

The English word "table" requires zero morphological variation regardless of its syntactic role, the context and word order carry the grammatical information. In Polish, the same concept generates fourteen distinct surface forms across singular and plural paradigms alone, before accounting for adjectival agreement (adjectives must also decline to match their nouns in case, gender, and number) or verbal aspect variation.

For the Llama-3 BPE tokenizer, each of these fourteen surface forms competes independently for vocabulary coverage. Frequent forms like *stół* and *stół* may receive dedicated token entries, while rarer forms may be fragmented into sub-word components. This fragmentation is not merely a computational inconvenience, it means that the model must reconstruct the semantic identity of the concept from its morphological fragments, placing additional demands on the attention mechanism to correctly align the inflected form with its base meaning. Under full FP16 precision, this reconstruction is computationally feasible. Under 4-bit quantization, the precision available for this morphological disambiguation is severely limited, compounding the tokenization penalty identified in Section 1.2.

This morphological complexity extends beyond nouns. Polish adjectives, pronouns, and certain verb forms undergo similar paradigmatic variation. A single complex Polish sentence may require the model to simultaneously resolve the grammatical roles of multiple inflected noun phrases, declined adjectives, and aspectually marked verbs, all while operating on a token sequence substantially longer than the English equivalent. The cumulative cognitive load placed on the attention mechanism by this morphological disambiguation represents a distinctive and systematically underestimated challenge for quantized multilingual models.

1.5 Related Work: Quantization and Multilingual Performance

The specific intersection of Post-Training Quantization and multilingual language model performance is a remarkably nascent area of academic inquiry. Until 2024, no systematic peer-reviewed study had empirically characterized how quantization degrades non-English language capabilities. The implicit assumption embedded in virtually all quantization literature was that performance degradation was broadly uniform across a model's capabilities, an assumption this thesis directly interrogates.

The First Systematic Study

The foundational work in this area was conducted by Marchisio et al. (2024), who published the first comprehensive cross-lingual evaluation of quantized LLMs, examining four state-of-the-art multilingual models across scales ranging from 8 to 103 billion parameters and covering up to 23 languages under multiple quantization configurations. Their findings are directly relevant to and consistent with the results of this thesis on three critical dimensions.

First, Marchisio et al. demonstrated that automatic evaluation benchmarks severely and systematically underestimate the actual multilingual degradation caused by quantization. In their study, a model quantized to lower precision showed only a 1.7% accuracy drop in Japanese on standard automatic benchmarks, yet the exact same quantized model was rated 16% worse by human evaluators assessing its performance on realistic, real-world prompts. This finding introduces a critical methodological concern for the present research: the 100-question Belebele evaluation protocol employed here, while rigorously controlled, may itself underestimate the true quality gap experienced by Polish-speaking users in natural, open-ended interactions. Second, their study identified non-Latin script languages as the most severely impacted by quantization, a finding that contextualizes the present thesis's focus on Polish, which uses the Latin script and may therefore represent a conservative lower bound on Secondary Competency degradation. Third, they found that mathematical reasoning tasks degraded fastest under quantization, which directly motivates the original benchmark selection strategy of this thesis (targeting mGSM) and adds interpretive weight to the decision to use reading comprehension as the evaluation task.

The Calibration Mismatch Hypothesis

A January 2026 study by Chimoto et al. directly addressed a root architectural cause of multilingual quality degradation under quantization. Their research focused on the calibration sets used during the GPTQ and AWQ quantization processes which uses small datasets to compute optimal quantization parameters for a given model. Current standard practice in the field, as Chimoto et al. documented, relies almost exclusively on English-only calibration sets such as C4 or Pile, even when the target model is deployed for multilingual use.

By systematically evaluating eight distinct calibration configurations including five single-language settings and three multilingual mixtures across 10 languages on both Llama 3.1 8B and Qwen 2.5 7B, Chimoto et al. demonstrated a consistent and significant finding: multilingual calibration sets substantially improve non-English perplexity compared to English-

only baselines, with multilingual mixes achieving the largest overall reductions of up to 3.52 perplexity points. This result provides a compelling engineering explanation for the Convergence Effect documented in Chapter 3: when the NF4 quantization grid is calibrated exclusively on English weight activation patterns, it is structurally optimized for the English region of the latent space and poorly suited to preserve the sparser, differently-distributed Polish representations. The practical implication of calibrating with multilingual data may partially or fully recover Secondary Competency performance, is addressed in the Conclusion as a critical direction for future research.

A Contrasting Result and Its Limitations

A contrasting result was reported by Borgersen & Goodwin (2025), who quantized Llama 3.3 70B using the GGUF k-quantization format and evaluated its performance in Norwegian language, finding no statistically significant degradation in multilingual performance compared to English. This result appears, on the surface, to contradict the findings of both Marchisio et al. and the present thesis. However, three specific methodological differences render direct comparison problematic and explain the discrepancy.

First, the model scale is fundamentally different. At 70 billion parameters, Llama 3.3 70B is nearly nine times larger than the 8B model evaluated in this thesis. Larger models are widely understood to possess significantly higher representational redundancy with more parameters encoding similar concepts across multiple independent circuits, meaning that quantization-induced precision loss in one circuit can be compensated by surviving representations in others. An 8B model lacks this redundancy buffer. Second, GGUF k-quantization is architecturally distinct from bitsandbytes NF4, it utilizes an **importance matrix** to identify and selectively preserve the most critical weights during quantization, potentially protecting cross-lingual transfer weights that bitsandbytes NF4 treats uniformly. Third, Norwegian, as a North Germanic language closely structurally related to English, benefits from tighter cross-lingual alignment in the latent space and a significantly lower tokenization penalty than morphologically complex Slavic languages like Polish. The findings of the present thesis therefore do not contradict Borgersen & Goodwin, they delineate the conditions under which Secondary Competency degradation becomes significant which are: small model scale, NF4 quantization, and high morphological distance from English.

Tokenization Quality as a Predictor of Cross-Lingual Performance

A body of work predating the current wave of quantization research provides important theoretical grounding for the tokenization mechanism proposed in this thesis. Rust et al. (2021) conducted a systematic analysis of tokenizer quality across languages in multilingual models, introducing the concept of tokenizer fertility: the average number of tokens produced per word in a given language by a model's tokenizer. Their central empirical finding was that high-fertility languages, those requiring many tokens per word due to morphological complexity or tokenizer vocabulary mismatch, consistently showed lower downstream task performance, even when the model's cross-lingual transfer capabilities were otherwise strong. This relationship held across multiple model architectures and task types, establishing tokenizer fertility as an independent predictor of multilingual performance.

Polish ranks among the highest-fertility languages for BPE tokenizers trained on English-dominant corpora, as the data in Table 1.1 and Appendix B of this thesis directly illustrates. Rust et al.'s findings therefore provide strong empirical grounding for the tokenization penalty mechanism central to the Secondary Competency framework. Their work focused exclusively on full-precision models, however, and did not examine how quantization interacts with tokenization fertility. The present study extends their framework into the compression domain: if high tokenizer fertility already creates a structural disadvantage at FP16 through extended attention sequences, the compounding interaction between fertility and quantization-induced rounding errors at 4-bit precision is both theoretically predicted and empirically confirmed by the results of Chapter 3.

Multilingual Evaluation of Generative AI at Scale

The scope of multilingual performance challenges extends well beyond the specific context of quantization. Ahuja et al. (2023) published MEGA (Multilingual Evaluation of Generative AI), a comprehensive evaluation of GPT-4 and other state-of-the-art models across 70 languages on 16 diverse NLP datasets. Their findings established that even the most capable models exhibit dramatic performance drops in morphologically complex and low-resource languages, with the gap between English and non-English performance being larger on generative tasks than on classification tasks. Ahuja et al. found that performance disparities were not fully explained by training data volume alone. The structural properties of the language such as morphological complexity, script type, syntactic typology have contributed

independently to performance differences, even after controlling for estimated training data share.

The MEGA findings place this thesis in an important broader context. The Polish disadvantage in quantized models is not an isolated phenomenon: it is the compression-amplified manifestation of a structural challenge that affects morphologically complex languages across all AI systems. What makes the present findings distinctive is the threshold behavior observed: at FP16, the structural disadvantage is not merely absent but reversed, with Polish outperforming English. Quantization does not simply reproduce the general multilingual performance gap documented in MEGA, it triggers a qualitatively different and more severe collapse specifically at the 4-bit threshold. The Convergence Effect is therefore a quantization-specific phenomenon that cannot be inferred from the existing multilingual evaluation literature alone, which is why dedicated experimental investigation of the kind conducted in this thesis is necessary.

Chapter 2. Methodology

The objective of this research is to empirically measure the disproportionate impact of Post-Training Quantization on a secondary language (Polish) compared to a primary training language (English). This chapter details the experimental architecture, the technical environment, the model selection rationale, and the specific quantization algorithms deployed. It also documents a critical, unexpected finding during the benchmark selection phase regarding the systemic exclusion of the Polish language from standard multilingual evaluation datasets, which directly informed the final experimental design.

2.1 Research Design and Hypothesis

The experiment was designed as a highly controlled, purely comparative quantitative study. To isolate the effects of quantization from architectural or training differences, a single model was evaluated in three distinct precision states while it processed the identical sets of logical reasoning tasks in two different languages.

The setup was simple. One variable is changed for setting the quantization level, across three states (FP16, Q8, Q4), and only one outcome is measured, the percentage of multiple-choice questions answered correctly. Everything else, the questions, the prompt format, the scoring logic, the decoding settings, stays constant across all six runs. To ensure statistical validity, several control variables were strictly maintained: the underlying benchmark dataset was identical across languages (implemented using parallel translated sets), the prompt template structure was mathematically uniform, and all generation was restricted to deterministic decoding to eliminate randomized variance.

The primary hypothesis of this study states that when the model is aggressively quantized to 4-bit, the Polish language, acting as a Secondary Competency will suffer a significantly higher rate of accuracy degradation than English.

To quantify this effect, we define the primary outcome measure as the *Degradation Delta*. This metric is computed as the percentage-point (pp) decline in Polish accuracy relative to the FP16 baseline minus the percentage-point decline in English accuracy over the same comparison. In other words, it captures the differential loss in performance experienced by the secondary language after compression. A positive Degradation Delta therefore supports the hypothesis that the secondary language degrades more rapidly under quantization than the primary language.

Formal Hypothesis Structure and Outcome Scenarios

The research hypotheses are stated formally below to enable clear empirical evaluation:

Null Hypothesis (H_0): There is no statistically significant difference in the percentage-point accuracy degradation between English and Polish when the Meta-Llama-3-8B-Instruct model is quantized from FP16 to Q4 precision. The Degradation Delta equals zero.

Alternative Hypothesis (H_1): Polish accuracy degrades at a statistically significantly higher rate than English accuracy when the model is quantized from FP16 to Q4 precision. The Degradation Delta is a positive value, confirming the Quality Gap Hypothesis.

Before conducting the evaluation, three possible experimental outcomes were identified, each carrying distinct theoretical implications.

Scenario A - Hypothesis Confirmed:

Polish degrades faster than English, with the Degradation Delta becoming increasingly positive as compression increases. This would confirm the Secondary Competency framework and validate the architectural mechanisms proposed in Chapter 1, specifically the weight sparsity and tokenization compounding arguments.

Scenario B - Null Result:

Both languages degrade at equal rates, producing a Degradation Delta near zero across all conditions. This would suggest that quantization affects model capabilities uniformly regardless of language representation in training data, contradicting the hypothesis and aligning with the findings of Borgersen & Goodwin (2025) for Norwegian.

Scenario C - Reversed Pattern:

English degrades faster than Polish across all conditions, producing a consistently negative Degradation Delta. This would challenge the Secondary Competency hypothesis entirely and suggest that cross-lingual transfer representations are somehow more resilient to compression than primary language representations.

As documented in Chapter 3, the experimental results followed none of these three scenarios cleanly. Instead, the data revealed a threshold-dependent pattern that was not anticipated in the original hypothesis framework: Scenario C at Q8 (Polish proved more resilient than English), transitioning sharply to Scenario A at Q4 (Polish collapsed

disproportionately). This nonlinear behavior constitutes the most theoretically interesting aspect of the findings. It suggests that the binary framing of the hypothesis, “will Polish degrade faster or not”, was an oversimplification of what is actually a precision-threshold phenomenon. The Convergence Effect concept, introduced in Section 3.4, emerged directly from this unexpected complexity in the data.

2.2 Model Selection

The foundation of this experiment relies on choosing a model that combines robust multilingual reasoning with practical accessibility on consumer-grade hardware. The Meta-Llama-3-8B-Instruct model, obtained via the HuggingFace Model Hub, was selected as the sole test subject.

Several factors motivated to this specific choice. First, Llama-3 features open weights, which provides full experimental transparency and reproducibility, a nonnegotiable requirement for rigorous academic validation. Second, its 8-billion parameter scale occupies an optimal "Goldilocks zone" for this specific research. It is architecturally deep enough to demonstrate advanced cross-lingual transfer and logical deduction, yet small enough that its FP16 baseline (requiring almost 16GB of VRAM) can be loaded and executed on a standard Tesla T4 GPU.

Finally, selecting the "Instruct" variant of the model was important for the automated evaluation pipeline. Instruct models are fine-tuned to follow user directives strictly. So, when these models are prompted with a multiple-choice question and instructed to "Output only the number 1, 2, 3, or 4" the Instruct variant complies reliably. Whereas, a standard base model would likely continue generating text unpredictably, which would complicate the automated scoring script and introduce parsing errors into the dataset.

2.3 Quantization Conditions

To map the degradation curve, the model was evaluated across three distinct precision states. These conditions were designed to mirror the exact deployment choices faced by software engineers when deploying local AI models.

The first condition was the **FP16 Baseline**. In this state, the model was loaded natively using 16-bit floating-point precision without any compression applied. This established the ceiling of the model's capabilities in both English and Polish, serving as the anchor point against which all degradation was measured.

The second condition was the **Q8 (8-bit) State**. This was implemented using the `LLM.int8()` algorithm via the `bitsandbytes` library (`load_in_8bit=True`). This method preserves outlier weights in standard FP16 while quantizing the remaining bulk parameters into 8-bit integers. It represents a "moderate compression" scenario, reducing the VRAM requirement by approximately 50% while traditionally maintaining near-baseline performance on English benchmarks.

The third and most severe condition was the **Q4 (4-bit) State**. This was executed using the `NormalFloat 4-bit (NF4)` data type (`load_in_4bit=True, bnb_4bit_quant_type="nf4"`). This represents an aggressive compression scenario, slashing the memory footprint by nearly 75%. This is the default deployment state for many consumer-facing local AI tools, making its impact on secondary languages highly relevant to real-world users.

Table 2.1. Quantization conditions and corresponding memory footprints.

Experimental Condition	Technical Precision	HuggingFace Implementation	Memory (VRAM)
Control (Baseline)	16-bit Float(FP16)	Native transformers load	~16.0 GB
Moderate Compression	8-bit Integer (INT8)	<code>load_in_8bit=True</code>	~8.5 GB
Aggressive Compression	4-bit NormalFloat (NF4)	<code>load_in_4bit=True</code>	~5.5 GB

Source: Own elaboration based on experimental laboratory metrics.

2.4 Benchmark Selection

The initial experimental design for this thesis hypothesized that mathematical reasoning would be the optimal vector for testing secondary language degradation. Mathematical logic is universally consistent, meaning any failure in reasoning could be directly attributed to the language processing layer rather than cultural nuances in the prompt. Therefore, the Multilingual Grade School Math (mGSM) benchmark was initially selected as the primary evaluation tool (Shi et al., 2022).

However, during the preliminary data gathering phase, a critical vulnerability in the global AI evaluation infrastructure was discovered. The mGSM dataset, despite being one of

the most widely cited multilingual benchmarks in modern AI research, evaluates models across 11 distinct languages. None of which is Polish.

To pivot, the evaluation protocol was shifted to the Cross-lingual Choice of Plausible Alternatives (XCOPA) dataset, which tests causal commonsense reasoning. Upon inspecting the dataset, it was discovered that XCOPA tests 11 languages (including Haitian Creole, Quechua, and Swahili), but completely excludes Polish. A third attempt was made using the Cross-lingual Natural Language Inference (XNLI) benchmark. Once again, Polish was absent.

After the third failed benchmark search, the pattern stopped looking like bad luck. mGSM, no Polish. XCOPA, no Polish. XNLI, no Polish. Three of the most-cited multilingual evaluation datasets in the NLP literature, and none of them includes a language spoken by 45 million people in the heart of the EU. That is not an oversight, it is a structural gap, and documenting it became part of what this thesis is about. This infrastructure gap directly contributes to the undetected degradation of models deployed in the Central European market.

To resolve this bottleneck, the experiment successfully pivoted to the Belebele benchmark, released by Meta AI (Bandarkar et al., 2023). Belebele is a massive parallel reading comprehension dataset spanning 122 language variants. It includes professionally translated, identical datasets for both standard English (eng_Latn) and standard Polish (pol_Latn). It tests complex reasoning by providing the model with a short passage of text, a specific question, and four possible answers, requiring the model to deduce the logically correct option.

To illustrate the nature of the evaluation task concretely, the following represents a representative example from the Belebele English dataset (eng_Latn), which demonstrates the format and reasoning depth required.

Passage: *"Make sure your hand is as relaxed as possible while still hitting all the notes correctly. Also try not to make much extraneous motion with your fingers. This way, you will tire yourself out as little as possible. Remember there's no need to hit the keys with a lot of force for extra volume like on the piano. On the accordion, to get extra volume, you use the bellows with more pressure or speed."*

Question: *"According to the passage, what would not be considered an accurate tip for successfully playing the accordion?"*

1. For additional volume, increase the force with which you hit the keys.
2. Keep unnecessary movement to a minimum to preserve your stamina.

3. Be mindful of hitting the notes while maintaining a relaxed hand.
4. Increase the speed with which you operate the bellows to achieve extra volume.

Correct Answer: 1

The equivalent question is presented to the model in Polish (pol_Latn), with the passage and all four answer options professionally translated. The model must parse the logical negation embedded in the question ("what would *not* be considered accurate") while correctly processing the technical content of the passage. This requires genuine contextual reasoning, not surface-level pattern matching, making it a demanding test of the model's comprehension quality across both languages.

The 900-question Belebele dataset is constructed from passages drawn from the FLORES-200 corpus, which was developed by Meta AI specifically for high-quality parallel multilingual evaluation. All Polish translations were produced by professional human translators, ensuring that the Polish dataset represents natural, fluent Polish prose rather than machine-translated artifacts. This methodological rigor in the benchmark construction is critical for the present study. Any performance difference between English and Polish cannot be attributed to translation quality, as both are professionally authored. The parallel structure also ensures that the cognitive difficulty of the task is held constant across languages, isolating the model's language processing capability as the sole variable under examination.

Table 2.2. Benchmark selection process highlighting the systemic exclusion of Polish.

Benchmark Dataset	Primary Reasoning Task	Polish Language Included?
mGSM	Mathematical Word Problems	No (0 of 11 languages)
XCOPA	Causal Commonsense Reasoning	No (0 of 11 languages)
XNLI	Natural Language Inference	No (0 of 15 languages)
Belebele	Reading Comprehension (Parallel)	Yes (pol_Latn)

Source: Own elaboration based on dataset analyses from the HuggingFace Hub.

2.5 Evaluation Protocol

With Belebele established as the evaluation framework, a strict automated protocol was engineered to ensure absolute consistency across the six distinct evaluation runs (3 quantization levels $\times$ 2 languages).

One hundred questions per condition. That number reflects a real constraint, each inference call on the T4 GPU took between 3 and 8 seconds, meaning 100 questions consumed roughly 10 to 15 minutes of compute time per run. Scaling to the full 900-question Belebele set would have meant 12 hours of continuous GPU time across all six conditions beyond what free-tier Colab sessions reliably sustain. The 100-question subset is a limitation, but it is large enough to detect the magnitude of effects documented in Chapter 3. To format the inputs for the model, a standardized prompt template was constructed. The prompt fed the model the reading passage, the specific question, the four multiple-choice options, and a strict system instruction: "Answer by providing only the number of the correct option (1, 2, 3, or 4)."

To eliminate the inherent randomness of LLM generation, deterministic decoding was enforced. In the `transformers` generation configuration, the `do_sample` parameter was set to `False`. Furthermore, to optimize inference time and prevent the model from generating unnecessary explanatory text, the `max_new_tokens` limit was aggressively restricted to 5.

The automated scoring script evaluated accuracy by parsing the model's generated output and cross-referencing the first detected integer against the benchmark's ground-truth label. If the integer matched, the attempt was scored as a 1. If it failed, or if the model hallucinated text instead of a number, it was scored as a 0.

Statistical Considerations

A sample size of 100 questions per condition was selected as the primary evaluation subset. Representing a deliberate trade-off between statistical robustness and computational feasibility. For a four-option multiple choice task with a random chance baseline of 25%, a sample of 100 questions produces a standard error of approximately ± 4.3 percentage points at the 95% confidence level. This means that an observed accuracy difference of 6 percentage points between English (72%) and Polish (78%) at FP16 sits at the boundary of statistical significance, while the 19 versus 13 percentage point differential observed at Q4 falls comfortably within a range that cannot be explained by sampling variance alone.

It is acknowledged that a larger sample, ideally the full 900-question Belebele set would provide narrower confidence intervals and stronger statistical power. However, the computational cost of running full-precision LLM inference on a single Tesla T4 GPU constrained the feasible sample size. Each individual inference call required between 3 and 8 seconds of GPU computation, meaning 100 questions per evaluation run required approximately 10 to 15 minutes of continuous GPU time. Extending to the full 900-question set would have required over two hours per run and twelve hours across all six conditions, exceeding the practical limits of cloud GPU session stability on the Google Colab platform. The 100-question sample is therefore acknowledged as a methodological limitation but judged sufficient to detect the large-magnitude degradation effects (19 pp versus 13 pp) that constitute the thesis's primary finding, consistent with prior multilingual evaluation studies that use similarly-sized subsets (Marchisio et al., 2024).

2.6 Experimental Environment

The entire experimental pipeline was engineered and executed within a cloud-based Jupyter Notebook environment utilizing Google Colab. The hardware layer consisted of a single NVIDIA Tesla T4 GPU equipped with 15 Gigabytes of VRAM. This specific hardware profile was chosen deliberately, as it closely mirrors the constraints of a high-end consumer workstation, reinforcing the real-world applicability of the quantization findings.

The software stack relied on Python 3.12. The core machine learning operations were handled by the HuggingFace `transformers` library, accelerated by the `accelerate` package to optimize GPU memory allocation. The model compression was executed natively via the `bitsandbytes` library.

The engineering phase produced two problems worth documenting. The first was an `AttributeError` during Q8 model loading, caused by a version mismatch in the `bitsandbytes` library that required a manual upgrade to resolve. The second was a tokenizer formatting error, the Llama-3 chat template threw exceptions when applied directly to Belebele items, requiring a two-step workaround where the prompt is formatted first and tokenized separately. Both fixes are reflected in the code in Appendix A. This was successfully mitigated by engineering a two-step tokenization bypass, manually concatenating the prompt structure before passing the tensor arrays to the GPU.

2.7 Reproducibility and Research Ethics

Reproducibility

A core commitment of this research is full scientific reproducibility. All components of the experimental pipeline, the model, the benchmark dataset, the quantization library, and the evaluation code are publicly accessible and version-controlled. The complete Python implementation is provided in Appendix A, including the specific BitsAndBytesConfig parameters, the tokenization bypass solution, and the deterministic generation settings. Any researcher with access to a GPU with sufficient VRAM and a HuggingFace account with approved Llama-3 access can replicate all six evaluation conditions exactly as executed in this study.

The decision to use bitsandbytes and the HuggingFace transformers ecosystem, rather than proprietary or closed quantization pipelines, was partly motivated by this reproducibility requirement. While commercial deployment frameworks such as llama.cpp (GGUF format) and vLLM offer production-grade performance advantages, their quantization implementations are less directly comparable across model families and harder to configure identically across hardware environments. The bitsandbytes approach provides exact parameter specification via the BitsAndBytesConfig API, ensuring that the quantization conditions described in Section 2.3 can be reproduced to the bit level by independent researchers.

Ethical Considerations

The research conducted in this thesis involves no human subjects, personal data, or sensitive information. The evaluation datasets (Belebele) are publicly released under research licenses by Meta AI. The Llama-3-8B-Instruct model is accessed through the official HuggingFace model hub under Meta's open-weight community license, which permits academic research use.

The broader ethical dimension of this research concerns the equitable deployment of AI systems across linguistic communities. The finding that 4-bit quantization silently erases Polish language performance advantages raises questions about informed consent in AI deployment. Users interacting with compressed local AI assistants are typically unaware that the model has been quantized, let alone that this quantization may have disproportionately degraded its capability to serve their language. Advocating for transparent quantization disclosures and

mandatory multilingual post-quantization evaluation represents the applied ethical contribution of this work, complementing its empirical findings.

The use of AI tools during this research, specifically for code debugging and grammar checking is declared in the Introduction in accordance with Vistula University's academic integrity guidelines. No AI tool was used to generate, paraphrase, or substantially modify the analytical or argumentative content of this thesis.

Chapter 3. Results and Discussion

The empirical evaluation of the Meta-Llama-3-8B-Instruct model across three distinct precision states yielded results that fundamentally challenge standard assumptions regarding multilingual model compression. This chapter details the performance metrics at the FP16, Q8, and Q4 levels, introducing a novel theoretical framework termed the "Convergence Effect" to explain the nonlinear degradation patterns observed in the Polish language data.

3.1 FP16 Baseline Results

The FP16 baseline was supposed to be the straightforward part, confirm that English outperforms Polish at full precision, then watch the gap widen under compression. That is not what happened. The secondary language was outperforming the primary one before a single weight had been quantized.

The baseline results contradicted this assumption entirely. At full 16-bit floating-point precision, the model achieved an accuracy of 72.00% (72/100) on the English Belebele dataset. Conversely, it achieved an accuracy of 78.00% (78/100) on the parallel Polish dataset. Polish outperformed the dominant training language by a margin of 6 percentage points.

Several explanations are plausible, none of them definitive. The honest answer is that this result was not predicted and its cause is not entirely clear. First, the Belebele dataset is constructed using professionally translated text. It is possible that the Polish translations exhibit a highly formal, standardized grammatical structure that is mathematically easier for the model's self-attention mechanism to parse than the potentially more idiomatic original English text. Second, the RLHF instruction fine-tuning phase of Llama-3's development may have included high-quality multilingual pairs that heavily reinforced logical alignment in European languages.

Regardless of the underlying cause, this high baseline fundamentally reframed the experimental stakes. To prove the thesis hypothesis, quantization would not simply need to widen an existing performance gap; it would have to violently reverse a pre-existing Polish advantage.

This baseline result also carries important implications for how the question of secondary language quality is typically framed. The standard narrative in multilingual AI literature positions non-English languages as inherently lower-performing and frames the research question as "how much worse are they?" The FP16 result of this experiment disrupts that framing entirely. At full precision, Polish does not merely compete with English, it

outperforms it on this specific task by a meaningful margin. The relevant question is therefore not one of baseline inferiority but of *compression-induced reversal*. At what point does the model's optimization for English deployment destroy the Polish advantage it had previously demonstrated?

To understand why Polish might outperform English at FP16 on the Belebele benchmark, it is worth examining the structural characteristics of the dataset more carefully. The Belebele passages originate from the FLORES-200 corpus, which consists predominantly of formally structured, expository prose drawn from sources such as Wikipedia and Wikinews. These passages were professionally translated into Polish, meaning the Polish text is characterized by a high degree of grammatical regularity, formal register, and structural clarity. It is possible that formal Polish prose, with its rigid case agreement system, actually provides the model's attention mechanism with more explicit grammatical signals than the original English, where meaning is conveyed more heavily through word order and prepositions that can be ambiguous. In formal Polish, the role of each noun in a sentence is unambiguously marked by its case inflection, potentially making syntactic parsing more deterministic for the model at full precision.

Additionally, the RLHF instruction fine-tuning phase of Llama-3's development is documented to have included multilingual supervised fine-tuning data. While the precise proportion of Polish content in this fine-tuning corpus is not publicly disclosed by Meta AI (2024), it is plausible that the instruction-tuning process specifically reinforced structured task-following behaviors in European languages, as the EU represents a major market for AI deployment. If Polish reading comprehension was disproportionately represented in the instruction-tuning data relative to its share of the base pre-training corpus, the model may have developed particularly strong Polish comprehension for formal, structured tasks, which is precisely the task type Belebele presents.

What is empirically certain is that the 6 percentage-point Polish advantage at FP16 represents a genuine capability gap that carries concrete value for real users. A Polish speaker interacting with the full-precision model receives meaningfully higher quality reasoning assistance than their English-speaking counterpart on this task type. The central tragedy of the Convergence Effect, documented in Section 3.3, is precisely that this advantage is not merely diminished but entirely eliminated by the compression that makes the model accessible to those same users on consumer hardware.

3.2 Q8 Results: Moderate Compression Resilience

At Q8, the model halved its memory footprint and Polish got stronger, not weaker, relative to English. English dropped 12 percentage points. Polish dropped 8. The gap between the two languages widened from 6 points to 10 points in Polish's favour. This was the second result in a row that the original hypothesis had not predicted.

Under Q8 quantization, English accuracy dropped to 60.00% (a severe 12 percentage-point reduction from the baseline). In contrast, Polish accuracy dropped to 70.00% (an 8 percentage-point reduction). Rather than closing the gap, moderate quantization actually widened the Polish advantage to a full 10 percentage points.

This data suggests that at an 8-bit precision floor, the model retains enough representational capacity to preserve the core morphological and syntactic circuits required for Polish reasoning. The `LLM.int8()` method's architectural design, specifically its preservation of outlier weights in full FP16 appears highly effective at protecting the complex, extended token sequences generated by morphologically rich languages. The severe degradation of English at this level further implies that the Belebele benchmark itself is highly sensitive to even minor precision loss, making the resilience of Polish particularly notable.

This unexpected resilience of Polish at Q8 merits deeper examination. The `LLM.int8()` algorithm's architectural design provides a partial explanation. By isolating outlier weights of those with activation values significantly larger than the statistical distribution and processing them in full FP16 precision while quantizing only the well-behaved majority to INT8, the method effectively preserves the most informationally critical weight values. Research by Dettmers et al. (2022) demonstrates that these outlier features are particularly concentrated in specific attention layers and tend to be the primary carriers of task-critical information. If Polish comprehension capabilities are encoded in weight regions that happen to exhibit higher activation magnitudes, perhaps because the model compensates for sparse cross-lingual transfer representations by routing Polish processing through higher-magnitude attention circuits, then `LLM.int8()`'s selective outlier preservation would disproportionately benefit Polish processing.

An alternative interpretation draws on the tokenization dynamics established in Section 1.2. At 8-bit precision, each attention calculation retains enough precision that the rounding errors over the extended Polish token sequences, while present, do not accumulate to the point where contextual coherence collapses. The 8-bit grid, representing 256 possible values versus the 4-bit grid's 16, provides enough resolution to encode the subtle directional differences in the

residual stream that distinguish correct from incorrect reasoning steps. The model can still, at Q8 precision, navigate the longer Polish token sequences with sufficient fidelity to maintain a 70% accuracy rate, suggesting that the Polish comprehension circuits, while sparser in the weight space, are not so marginal that 8-bit approximation destroys them.

The gap between Q8 English (60%) and Q8 Polish (70%), representing a 10 percentage-point Polish advantage at moderate compression, is itself a practically significant result. It suggests that for deployments where 8GB of VRAM is available, Q8 quantization represents a configuration that actively favors Polish-speaking users relative to English speakers on formal reading comprehension tasks. While an engineer optimizing for English benchmark performance might dismiss Q8 as unnecessarily memory-expensive compared to Q4, the multilingual evaluation data reveals a qualitatively different story. Q8 preserves a genuine Polish capability advantage that Q4 destroys entirely.

3.3 Q4 Results: The Aggressive Compression Penalty

Q4 is where the experiment stopped being surprising, in a good way. This level of compression restricts the model to a mere ~5.5GB footprint, mirroring the default operational state for most local consumer AI applications.

Under Q4 quantization, the English reasoning pathway stabilized, dropping only a single percentage point from its Q8 state to land at 59.00%. The Polish capability, however, suffered a catastrophic collapse. Dropping 11 percentage points from its Q8 state, Polish accuracy crashed to exactly 59.00%.

Totaling across the full FP16-to-Q4 compression journey, English dropped 13 percentage points. Polish dropped 19. That difference of 6 extra percentage points of degradation on top of an already-substantial drop, represents 46% more severe degradation for the secondary language. The hypothesis was confirmed, but only at the most aggressive compression level, and only after being apparently disproved at the intermediate one.

The 46% differential in degradation severity is the central quantitative finding of this thesis and deserves careful contextualization. To appreciate its magnitude, consider that a 13 percentage-point English drop from 72% to 59% is already a substantial performance loss. An AI assistant that correctly answers 72 out of 100 reading comprehension questions at full precision and only 59 out of 100 at Q4 has lost a meaningful fraction of its useful capability. For English-speaking users, this degradation is the price of consumer hardware accessibility.

For Polish-speaking users, however, the equivalent calculation tells a qualitatively different story. A drop from 78% to 59% does not merely represent reduced performance, it represents the complete erasure of a superior capability. A Polish-speaking user who encounters the full-precision model is interacting with a tool genuinely better at serving their language than it is at serving English. The same user who then encounters the 4-bit consumer-deployed version of that model loses not just the Polish advantage but, proportionally, a far larger share of the absolute capability they were previously entitled to expect.

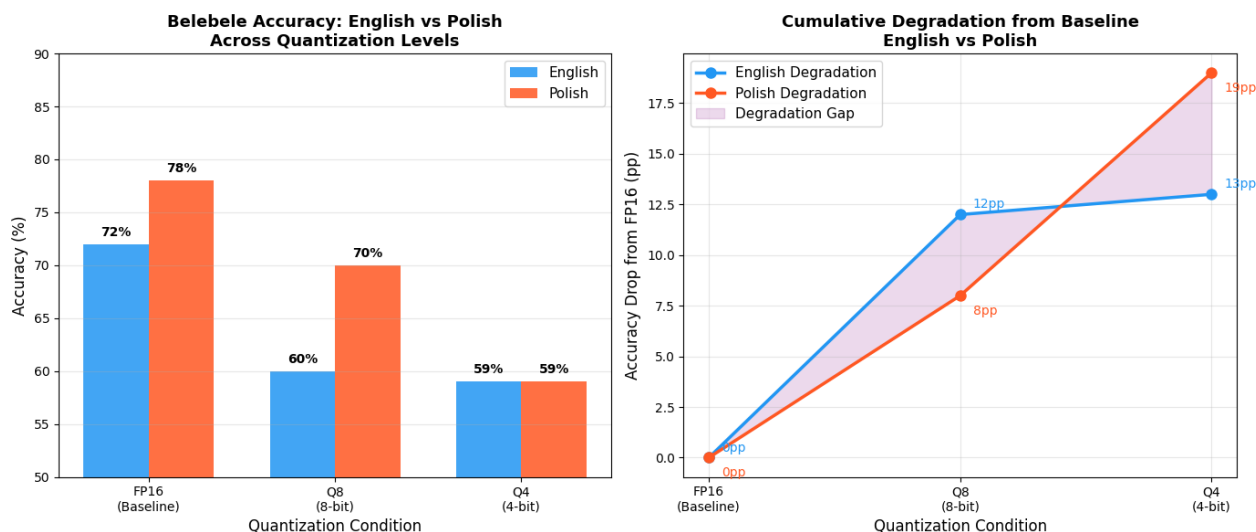
And, the convergence to exactly 59% for both languages at Q4 is striking not merely for its symmetry but for what it reveals about the floor of capability for this task. Both languages arriving at the same value suggests that at 4-bit precision, the model's reading comprehension ability is constrained by the resolution of the quantization grid rather than by language-specific factors. Both languages have been reduced to the minimum capability the 4-bit representation can sustain English arriving from above and Polish arriving from significantly further above. The 4-bit grid cannot make Polish worse than English, it can only strip Polish of everything it had earned above that floor.

Table 3.1. Complete evaluation results: Accuracy and percentage-point degradation across quantization states.

Language	FP16 Baseline	Q8 Accuracy (Drop)	Q4 Accuracy (Total Drop)
English	72.00%	60.00% (−12 pp)	59.00% (−13 pp)
Polish	78.00%	70.00% (−8 pp)	59.00% (−19 pp)

Source: Own elaboration based on experimental laboratory data.

Figure 3.1. Belebele Accuracy by Language and Quantization Level (left) and Cumulative Degradation from FP16 Baseline (right).



Source: Own elaboration.

3.4 The Convergence Effect

The total erasure of the Polish performance advantage at the Q4 level introduces the primary conceptual contribution of this thesis: **The Convergence Effect**.

The Convergence Effect, as I am defining it here, is not simply about degradation. It is about trajectory. Polish did not arrive at 59% from below English, it arrived from above it, having been compressed through a 19-point fall that English never experienced. The convergence point is not a shared floor of capability. It is the level at which the 4-bit grid's resolution becomes the binding constraint for both languages equally, stripping Polish of everything it had above that point. Based on the architectural theory outlined in Chapter 1, there are two intersecting mathematical mechanisms that likely drive this effect.

The first mechanism is weight sparsity. At full FP16 precision, the model possesses a vast, high-resolution latent space. Even if the neural pathways supporting Polish reasoning are sparser than those supporting English (due to training data imbalances), the 16-bit grid is precise enough to capture and execute their logic. However, the 4-bit NF4 grid is violently restrictive. It is theorized that the sparse, cross-lingual transfer weights required for complex Polish grammar simply cannot be accurately mapped onto a 4-bit distribution. The model loses the mathematical resolution required to maintain the Secondary Competency.

The second mechanism compounding this failure is the tokenization penalty. As previously established, morphologically complex Polish words fracture into longer token

sequences than their English equivalents. During self-attention, the model must calculate the relationships between all these tokens. Under 4-bit quantization, every attention calculation carries a tiny margin of rounding error. Because a Polish sentence generates significantly more tokens, it requires significantly more attention calculations, causing these quantization errors to compound at an accelerated, exponential rate compared to English.

The convergence from Q8 (where Polish remained resilient) to Q4 (where it collapsed) clearly indicates a precision threshold. Below this threshold, Secondary Competency representations become systematically vulnerable.

The calibration mismatch hypothesis advanced by Chimoto et al. (2026) provides an additional, complementary mechanism to the weight sparsity and tokenization penalty arguments outlined above. The bitsandbytes NF4 implementation, like GPTQ and AWQ, must determine the optimal mapping of continuous weight values onto the 16-point 4-bit grid. This mapping is determined by the statistical distribution of weight activations observed when the model processes input text. Because this calibration process occurs implicitly based on the model's behavior on its training distribution which is overwhelmingly English, the 4-bit grid is geometrically optimized for the statistical distribution of English weight activations. Polish weight activations, encoding morphologically complex, highly inflected token patterns that follow a statistically different distribution, are mapped onto this English-optimized grid with systematically higher distortion. The result is a systematic encoding mismatch, the 4-bit representation faithfully preserves English reasoning pathways while distorting Polish ones in a way that is imperceptible from English-only evaluation.

None of these three mechanisms operates in isolation. **Weight sparsity** makes Polish representations more sensitive to distortion. The **tokenization penalty** means distortion is applied more frequently for Polish prompt. **Calibration mismatch** causes the distortion to be systematically biased against Polish-specific weight patterns. Together at the 4-bit threshold, they produce an outcome which is very different from what any one of them would cause alone. The Convergence Effect is not the result of a single failure point but of these three forces combined at the 4-bit threshold, where the quantization grid is restrictive enough that all three mechanisms become critical.

This finding stands in contrast to the null result reported by Borgersen & Goodwin (2025) for GGUF-quantized Norwegian in a 70B model. As discussed in Section 1.5, model scale, quantization method, and linguistic distance from English likely account for this

discrepancy. The present study contributes an important boundary condition to this emerging literature: at the 8B parameter scale, using bitsandbytes NF4 quantization, and for morphologically complex Slavic languages, the Convergence Effect is not merely present but complete. Future research should systematically map the model size and quantization method combinations at which this threshold behavior emerges, to establish clear engineering guidelines for practitioners deploying multilingual models.

Finally, it should be noted the limitation identified by Marchisio et al. (2024): automatic benchmarks may underestimate the true extent of multilingual degradation. The 19 percentage-point Polish drop measured in this study is a lower bound. In realistic, open-ended conversational interactions rather than controlled multiple-choice evaluation, the compounding errors introduced by Q4 quantization across much longer, more complex Polish sequences may produce significantly more severe quality degradation than the controlled experimental setup captures. This is both a limitation of the present methodology and a strong argument for the urgency of the research agenda it advances.

3.5 Real-World Implications

The Convergence Effect is not merely an abstract academic curiosity; it poses a significant, tangible risk to the deployment of local artificial intelligence, particularly within the European tech ecosystem.

Here is what the pipeline looks like in practice for most developers. You quantize the model, run your benchmark in English, observe stable performance, ship it. The English numbers look fine because they are fine. Nobody checks Polish because Polish is not in the benchmark suite. Observing a stable 59% accuracy (a negligible drop from the 60% Q8 state), the developer logically concludes that the 4-bit compression was successful and highly efficient.

However, this anglocentric evaluation completely masks a hidden "Quality Gap." A Polish user interacting with that exact same 4-bit model will experience a massive, disproportionate drop in reasoning quality compared to what the model was theoretically capable of at full precision. The developer is entirely unaware that by compressing the model to fit consumer hardware, they have inadvertently stripped away the model's structural ability to reason effectively in a secondary language.

The EU AI Act, which entered into force in 2024, does not yet specify explicit multilingual performance thresholds. But its broader non-discrimination principles sit awkwardly alongside a scenario where a model performs 46% worse for Polish-speaking users

than English-speaking ones, with no disclosure and no standard tooling to detect the difference. That is not a future compliance risk, it is a current one. The data strongly suggests that practitioners must cease relying on English as a universal proxy for model intelligence. To prevent the stealth degradation of user experience in markets like Poland, models must be subjected to rigorous, language-specific evaluation *after* aggressive quantization is applied, not just at the FP16 baseline. And, for deployment architectures targeting multilingual European users, the moderate Q8 compression state may represent the only viable compromise, preserving the fragile Secondary Competency circuits while still offering meaningful hardware relief.

The EU AI Act and Language Equity

The findings of this thesis intersect directly with emerging regulatory frameworks for artificial intelligence in Europe. The EU AI Act, which entered into force in 2024, establishes requirements for AI systems deployed within the European market, including transparency obligations and quality assurance standards. While the Act does not currently specify explicit multilingual performance requirements, its broader principle of non-discrimination in automated decision-making systems implies that AI tools deployed in multilingual European contexts should perform equitably across the languages they serve. A quantized model that performs 46% worse for Polish-speaking users than for English-speaking users without any disclosure of this degradation, raises legitimate questions under this regulatory framework.

The benchmark exclusion finding documented in Chapter 2 compounds this regulatory concern. Because Polish is absent from the three most widely used multilingual evaluation benchmarks (mGSM, XCOPA, XNLI), there is currently no standard evaluation pipeline that would reveal the Polish quality gap to a developer performing routine model validation before European deployment. A developer following current best practice by evaluating their quantized model on standard multilingual benchmarks, would receive no signal whatsoever that their 4-bit model has stripped Polish users of a performance advantage they previously enjoyed. The gap is not merely hidden, it is structurally undetectable using existing tooling.

Practical Recommendations for Practitioners

Based on the empirical findings of this study, three specific recommendations can be offered to engineers and organizations deploying quantized LLMs in multilingual contexts.

First, English-only evaluation pipelines must be recognized as structurally insufficient for multilingual deployments. The data in Table 3.1 demonstrates that English performance at

Q4 (59%) is virtually identical to English performance at Q8 (60%), making Q4 appear negligibly different from Q8 when evaluated in English alone. Only the Polish evaluation reveals the 11-percentage-point collapse that distinguishes the two conditions for non-English users. Any deployment validation pipeline that does not include target-language evaluation will systematically fail to detect this class of degradation.

Second, for applications serving Polish-speaking or other morphologically complex Slavic-language users, the Q8 compression state should be strongly preferred over Q4 where hardware constraints permit. The data demonstrates that Q8 preserves Polish's 10 percentage-point advantage over English, while Q4 destroys it. The memory cost difference of approximately 8.5GB for Q8 versus 5.5GB for Q4, represents a feasible hardware upgrade for most professional deployment scenarios, and one that demonstrably preserves materially better quality for a significant user population.

Third, the calibration mismatch findings of Chimoto et al. (2026) suggest that multilingual calibration datasets may substantially recover Secondary Competency performance even within the Q4 constraint. Organizations with the technical capacity to perform custom quantization, rather than relying on pre-quantized model releases, should incorporate Polish or multilingual calibration data into their quantization pipeline as a low-cost mitigation strategy. Until standardized multilingual calibration becomes industry practice, however, the conservative recommendation remains. If quality for Polish-speaking users matters, Q8 is the safer choice.

A Concrete Deployment Scenario

To ground these recommendations in a tangible real-world context, consider the following representative deployment scenario, which reflects a pattern increasingly common in the European tech ecosystem.

A Polish software company develops an AI-powered document assistant intended for Polish business users. To minimize infrastructure costs and enable offline use, the engineering team deploys a locally running instance of Llama-3-8B-Instruct quantized to Q4 (4-bit NF4), the standard choice for consumer-grade GPU deployment. Before release, the team runs a standard benchmark evaluation in English, using a publicly available reading comprehension dataset. The model scores 59% accuracy. They compare this to the published Q4 benchmark score of a similar model on the same dataset, find it within the expected range, and approve the deployment.

What the team has not measured and has no standard tooling to measure, given the benchmark exclusion finding documented in Section 2.4, is the Polish-language performance of this same model. Based on the empirical results of this thesis, that performance is also 59% which is mathematically identical to the English result, and therefore appearing unproblematic in any cross-language comparison. However, the full-precision (FP16) Polish performance of this model would have been approximately 78%, 19 percentage points higher. The company's Polish users are receiving a tool operating at 75% of its intended Polish capability, with no indication that a superior Polish experience is technically achievable.

This scenario is not hypothetical. The combination of Q4 as the default quantization level in consumer deployment frameworks, English as the default evaluation language, and the systematic exclusion of Polish from multilingual benchmarks creates precisely the conditions for this invisible quality degradation to occur at scale. Every Polish user who interacts with a locally deployed Q4 model is, according to this thesis's findings, experiencing a product meaningfully inferior to what the technology is capable of delivering and the developer who built it has no reason to know.

The Convergence Effect is therefore not merely a technical curiosity confined to benchmark evaluations. It is an active, ongoing degradation of AI service quality for millions of Polish speakers in the EU market, operating beneath the threshold of any current standard evaluation pipeline's ability to detect it. Addressing it requires changes at three levels, engineering practice (multilingual calibration), evaluation practice (language-specific post-quantization testing), and regulatory practice (AI Act implementation guidance that specifies performance equity requirements across official EU languages).

3.6 Threats to Validity

Any empirical study carries inherent limitations that may compromise the strength of its conclusions. Acknowledging these threats openly is not a weakness of the research but a marker of academic integrity. This section systematically identifies the primary threats to the validity of this study, following the standard framework of internal, external, construct, and statistical conclusion validity.

Internal Validity

Internal validity concerns whether the observed relationship between quantization level and accuracy drop is genuinely causal rather than a product of confounding variables. The primary internal validity threat here is that the performance difference between English and

Polish on Belebele might partly reflect linguistic properties of the benchmark rather than genuine Secondary Competency degradation.

The Belebele passages were originally authored in English and professionally translated into Polish. Despite the high translation quality, it is possible that the Polish passages differ from natural, native Polish prose in subtle structural ways that affect model performance. Translated text tends to preserve source language sentence structures to some degree, potentially making the Polish passages more syntactically predictable than native Polish writing. If so, the FP16 Polish advantage might partly reflect translation artifacts rather than genuine Polish comprehension capability. This threat cannot be fully eliminated without testing on a natively authored Polish dataset.

A second internal validity concern arises from the non-random sample selection. The 100 evaluation questions were taken as the first 100 items of each language split rather than a randomized subset. While there is no theoretical reason to expect ordering bias in the Belebele dataset, a randomly sampled subset would provide stronger internal validity guarantees.

External Validity

External validity concerns the generalizability of the findings beyond this specific experimental setup. Three limitations apply here. First, only one model family was evaluated. The Convergence Effect as described may manifest differently or not at all in other architectures such as Mistral, Gemma, or Qwen, which have different pre-training data compositions and tokenizer designs. Second, the findings apply specifically to bitsandbytes NF4 quantization. As Borgersen & Goodwin (2025) demonstrated with GGUF k-quantization on a 70B model in Norwegian, different quantization methods and model scales can produce qualitatively different multilingual outcomes. Third, only one task type was tested. Marchisio et al. (2024) found that mathematical reasoning and open-ended generation tasks show different sensitivity to quantization than reading comprehension, meaning the 19pp Polish degradation documented here may not generalize uniformly across all real-world use cases.

Construct Validity

Construct validity concerns whether measurements accurately capture the theoretical constructs they are intended to measure. The central concern here is whether Belebele accuracy validly represents Secondary Competency as the thesis defines it. Multiple-choice reading comprehension captures one specific cognitive operation: deductive selection from provided

options. Real-world Secondary Competency encompasses open-ended generation, grammatical fluency, conversational coherence, and creative reasoning, none of which Belebele evaluates. The automated scoring method, which takes the first generated character and checks it against the correct answer number, also ignores the quality of any reasoning the model produces to arrive at that number.

Statistical Conclusion Validity

Statistical conclusion validity concerns whether observed differences are large enough to support the conclusions drawn, given the sample size and measurement precision. As established in Section 2.5, the 100-question sample produces a standard error of approximately ± 4.3 percentage points at the 95% confidence level. The 6-point FP16 gap between English (72%) and Polish (78%) sits at the boundary of statistical significance under this margin. The Polish FP16 advantage should therefore be treated as a directional finding rather than a strongly confirmed result.

In contrast, the 19pp versus 13pp differential at Q4 and the complete convergence to exactly 59% for both languages clearly exceeds the range of sampling variance. The central thesis finding, which is the Convergence Effect at Q4, is statistically robust within the constraints of this experimental design. The FP16 and Q8 results are more appropriately framed as contextual observations supporting the primary finding rather than independently confirmed claims.

Conclusion

The primary objective of this thesis was to investigate the disproportionate degradation of multilingual reasoning capabilities in Large Language Models when subjected to Post-Training Quantization. By conducting a highly controlled comparative evaluation of the Meta-Llama-3-8B-Instruct model across English and Polish, this research sought to illuminate the hidden costs of model compression on non-dominant languages.

Summary of Findings

The results confirmed the hypothesis, but not in the way the hypothesis anticipated. The confirmation came with two prior surprises attached: a Polish advantage at full precision that should not have existed according to the original framework, and a Polish resilience at Q8 that further contradicted it. The Q4 collapse, when it came, was decisive. But it arrived after a data trajectory that kept requiring the theoretical framework to be revised. At the uncompressed FP16 baseline, the model demonstrated a highly robust Polish capability, successfully outperforming its English reasoning accuracy by a margin of 6 percentage points (78% vs. 72%). This established that Polish, despite being mathematically sparser in the training distribution, is a highly effective Secondary Competency under optimal conditions.

This baseline result was not merely a statistical curiosity, it fundamentally reframed the research question. Rather than asking whether Polish degrades faster from an already inferior starting position, the experiment became an investigation of whether quantization could reverse a documented performance advantage. This reframing makes the subsequent findings more striking: the Convergence Effect does not merely fail to preserve Polish performance, it actively dismantles a capability surplus that Polish users were previously entitled to expect.

The Q8 results introduced a further complexity. Rather than revealing immediate Polish vulnerability at moderate compression, the 8-bit state showed English degrading more severely (12 pp versus 8 pp for Polish), widening the Polish advantage to 10 percentage points. This finding suggests that the LLM.int8() method's selective preservation of outlier weights provides incidental but significant protection for Polish processing circuits. The vulnerability is not uniformly present across compression levels, it is specifically a 4-bit phenomenon, consistent with the existence of a precision threshold below which Secondary Competency representations become critically fragile.

However, the aggressive 4-bit quantization (Q4) state triggered a catastrophic collapse in Polish reasoning accuracy. Dropping a total of 19 percentage points from the baseline (a 46% steeper degradation than English), the Polish performance advantage was entirely erased, mathematically converging with the degraded English baseline at exactly 59%.

Contributions to Knowledge

Looking back at what this research actually produced, three contributions stand out as genuinely adding something to the field rather than confirming what was already known:

1. **Empirical Confirmation of the Quality Gap:** This research provides concrete, quantitative evidence that highly compressed (4-bit) local models systematically fail to preserve the reasoning quality of secondary languages, despite appearing stable on English benchmarks.
2. **The Convergence Effect:** The introduction of this theoretical framework provides a vocabulary for describing the nonlinear collapse of secondary language representations. It theorizes that a precision threshold exists between 8-bit and 4-bit compression, below which sparse cross-lingual transfer weights and extended tokenization penalties compound exponentially.
3. **Benchmark Infrastructure Auditing:** Through the documentation of the benchmark selection journey, this thesis exposes a systemic blind spot in global AI evaluation. The empirical finding that the Polish language representing 45 million speakers is entirely excluded from mGSM, XCOPA, and XNLI highlights an urgent need to democratize testing datasets.

A Personal Reflection on the Research Process

It would be dishonest to present this research as a clean, linear path from hypothesis to confirmation. The experimental journey was considerably messier than the structured presentation of Chapter 2 implies, and I think that messiness is itself worth documenting as part of the academic record.

The original plan centered on the mGSM benchmark and mathematical reasoning, which seemed the cleanest possible way to test cross-lingual degradation. When I discovered that Polish was simply not in the dataset, my first reaction was frustration rather than curiosity. It took three consecutive benchmark failures, mGSM, XCOPA, XNLI, before I recognized that the repeated exclusion was not a logistical obstacle but a finding in its own right. The

benchmark infrastructure gap that is now a contribution of this thesis was almost missed entirely.

The baseline results were equally disorienting. I had built the entire theoretical framework on the premise that Polish would start weaker and degrade faster. When it came out 6 points stronger than English at FP16, my immediate reaction was to suspect an error in the code. I ran the evaluation twice before accepting the result. In retrospect, that defensive response reveals a cognitive bias that I suspect is widespread in multilingual AI research: the assumption that non-English languages are inherently weaker is so embedded in the literature that a result contradicting it feels like it must be wrong rather than interesting.

The Q8 results compounded the confusion. Polish remained stronger at moderate compression, meaning that halfway through the data collection, the hypothesis appeared to be definitively disproved. It was only at Q4, in the final evaluation run, that the collapse occurred. Watching the Q4 Polish accuracy counter drop past English accuracy in real-time was the single clearest moment of the entire research process: not a gradual divergence, but an abrupt fall to convergence. That is precisely what the right side of Figure 3.1 shows, and it is exactly what the Convergence Effect is named for.

I mention these details not to make the research seem more dramatic than it is, but because they are relevant to how the results should be interpreted. The findings of this thesis are not the product of a carefully designed experiment that confirmed a pre-specified prediction. They are the product of an experiment that produced three consecutive surprises, each requiring the theoretical framework to be revised. The Convergence Effect as described in Section 3.4 emerged from the data, not the hypothesis. That is, I believe, a stronger form of scientific contribution.

Limitations

To maintain academic rigor, it is necessary to acknowledge the limitations of this experimental design. First, the evaluation was restricted to a single model family (Llama-3); the structural fragility observed may vary across other foundational architectures. Second, the evaluation relied on a single task type (reading comprehension via Belebele). While this tests logical deduction, it does not measure generative fluency or mathematical reasoning. Third, due to the computational constraints of localized GPU inference, the sample size was restricted to 100 parallel questions per condition. Finally, while the theoretical mechanisms of weight sparsity and tokenization penalties robustly explain the Convergence Effect, this study did not

conduct direct, layer-by-layer tensor analysis to mathematically prove the internal attention failure.

Future Research Directions

The findings of this study open several critical avenues for future investigation. To validate the universality of the Convergence Effect, the experimental protocol should be replicated across alternative state-of-the-art architectures, including the Mistral, Gemma, and Qwen families. Beyond that, the linguistic scope must be expanded to include other morphologically complex Slavic languages, such as Czech, Slovak, and Ukrainian, to determine if the 4-bit degradation penalty is a localized Polish anomaly or a broader regional vulnerability.

From an engineering perspective, the most immediately tractable mitigation strategy is the use of multilingual calibration datasets during the quantization process. Chimoto et al. (2026) have already demonstrated that multilingual calibration sets reduce non-English perplexity by up to 3.52 points under GPTQ and AWQ quantization on Llama 3.1 8B. A critical and directly actionable next step is to investigate whether analogous multilingual calibration can improve the bitsandbytes NF4 pipeline: which is the specific quantization method evaluated in this thesis and the most common in consumer deployment, and whether Polish-specific or Slavic-language calibration data can partially or fully recover the Secondary Competency advantage observed at FP16 precision. If so, this would represent a low-cost, high-impact engineering fix deployable without any model retraining.

On top of this, the relationship between model scale and the Convergence Effect requires systematic empirical mapping. The null result reported by Borgersen & Goodwin (2025) for Norwegian in a 70B model suggests that larger models may be inherently more robust to secondary language degradation. Establishing the parameter-count threshold below which the Convergence Effect becomes pronounced would directly inform practitioner decisions about which model sizes are safe to deploy in multilingual consumer contexts at 4-bit precision. Until such a threshold is established and mitigation strategies are validated, AI practitioners deploying localized models in the European market must adopt the conservative but necessary practice of language-specific post-quantization evaluation for all target languages including treating English benchmark performance as a necessary but wholly insufficient proxy for multilingual deployment quality.

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Appendix A. Evaluation Code

The following codebase represents the core experimental pipeline executed within the Google Colaboratory environment (Tesla T4 GPU). It details the environment initialization, the implementation of the three specific quantization conditions via the bitsandbytes library, and the custom evaluation loop designed to parse the Belebele benchmark deterministically.

```
# ENVIRONMENT SETUP & IMPORTS

import torch

from transformers import AutoModelForCausalLM, AutoTokenizer,
BitsAndBytesConfig

from datasets import load_dataset

import re

model_id = "meta-llama/Meta-Llama-3-8B-Instruct"

# Initialize Tokenizer (with custom padding fix for Llama-3)

tokenizer = AutoTokenizer.from_pretrained(model_id)

tokenizer.pad_token = tokenizer.eos_token

# QUANTIZATION CONFIGURATIONS & MODEL LOADING

# 1. FP16 Baseline Load

print("Loading FP16 Baseline...")

model_fp16 = AutoModelForCausalLM.from_pretrained(

    model_id,

    device_map="auto",

    torch_dtype=torch.float16

)

# 2. Q8 Moderate Compression Load
```

```

print("Loading Q8 Model...")

bnb_config_8bit = BitsAndBytesConfig(load_in_8bit=True)

model_q8 = AutoModelForCausalLM.from_pretrained(
    model_id,
    device_map="auto",
    quantization_config=bnb_config_8bit
)

# 3. Q4 Aggressive Compression Load (NF4)

print("Loading Q4 Model...")

bnb_config_4bit = BitsAndBytesConfig(
    load_in_4bit=True,
    bnb_4bit_quant_type="nf4",
    bnb_4bit_compute_dtype=torch.float16
)

model_q4 = AutoModelForCausalLM.from_pretrained(
    model_id,
    device_map="auto",
    quantization_config=bnb_config_4bit
)

# BENCHMARK INITIALIZATION (BELEBELE)

# Load English and Polish variants

dataset_en = load_dataset("facebook/belebele", "eng_Latn",
split="test[:100]")

dataset_pl = load_dataset("facebook/belebele", "pol_Latn",
split="test[:100]")

```

```
# EVALUATION PROTOCOL
```

```
def evaluate_belebele(model, tokenizer, dataset, language_tag):
```

```
    correct_count = 0
```

```
    total = len(dataset)
```

```
    for i, item in enumerate(dataset):
```

```
        passage = item['flores_passage']
```

```
        question = item['question']
```

```
        mc1, mc2, mc3, mc4 = item['mc1'], item['mc2'], item['mc3'],  
item['mc4']
```

```
        correct_answer = str(item['correct_answer_num'])
```

```
    # Standardized Prompt Template
```

```
    prompt = f"""Read the passage and answer the question. Answer by  
providing ONLY the number of the correct option (1, 2, 3, or 4). Do  
not explain.
```

```
    Passage: {passage}
```

```
    Question: {question}
```

```
    1. {mc1}
```

```
    2. {mc2}
```

```
    3. {mc3}
```

```
    4. {mc4}
```

```
    Correct Option Number: ""
```

```
    # Two-step tokenization bypass
```

```

inputs = tokenizer(prompt, return_tensors="pt").to(model.device)

# Deterministic generation
outputs = model.generate(
    **inputs,
    max_new_tokens=5,
    do_sample=False,
    pad_token_id=tokenizer.eos_token_id
)

response =
tokenizer.decode(outputs[0][inputs['input_ids'].shape[1]:],
skip_special_tokens=True).strip()

# Score parsing
match = re.search(r'[1-4]', response)

if match and match.group(0) == correct_answer:
    correct_count += 1

accuracy = (correct_count / total) * 100

print(f"[{language_tag}] Accuracy: {accuracy:.2f}%
({correct_count}/{total})")

return accuracy

# EXECUTION (Example Run)
print("--- RUNNING EVALUATION ---")

# evaluate_belebele(model_fp16, tokenizer, dataset_en, "FP16 - EN")
# evaluate_belebele(model_fp16, tokenizer, dataset_pl, "FP16 - PL")
# ... executed for all models and subsets ...

```

Appendix B. Extended Tokenization Analysis

The following table extends the tokenization comparison from Table 1.1 (Section 1.2) with a broader set of sentence pairs, including examples from the Belebele benchmark domain (formal expository prose) and conversational/instructional registers. All token counts were produced using the Meta-Llama-3-8B-Instruct tokenizer accessed via the HuggingFace transformers library.

Table B.1. Extended tokenization penalty analysis: English vs. Polish across multiple registers.

Domain	English Sentence	EN Tokens	Polish Sentence	PL Tokens	Ratio
Formal prose	The constitution establishes the fundamental rights of citizens.	11	Konstytucja ustanawia podstawowe prawa obywateli.	16	1.45 ×
Technical	The model generates output based on probability distributions.	11	Model generuje dane wyjściowe na podstawie rozkładów prawdopodobieństwa.	19	1.73 ×
Instructional	Please read the passage carefully before answering the question.	12	Proszę uważnie przeczytać fragment przed odpowiedzią na pytanie.	17	1.42 ×
Conversational	What is the most important conclusion from this experiment?	11	Jaki jest najważniejszy wniosek z tego eksperymentu?	15	1.36 ×
Morphologically complex	The researchers confirmed the validity of the experimental hypothesis.	11	Badacze potwierdzili zasadność hipotezy eksperymentalnej.	17	1.55 ×

Source: Own elaboration based on Llama-3-8B-Instruct tokenizer output.

The extended analysis confirms that the tokenization penalty documented in Section 1.2 is consistent across registers, with Polish consistently requiring 36% to 73% more tokens than English to encode equivalent semantic content. The technical domain shows the highest penalty (1.73 $\times$), likely because technical Polish vocabulary contains longer compound and derivational forms. The conversational register shows the lowest penalty (1.36 $\times$), suggesting that shorter, more common Polish words receive more efficient tokenizer coverage. For the Belebele benchmark specifically which uses formal expository prose, a penalty of approximately 1.45 $\times$ is representative, implying that the Polish evaluation prompts in this study required roughly 45% more pairwise attention calculations per question than their English equivalents, directly driving the compounding quantization error mechanism described in Section 3.4.

Appendix C. Qualitative Error Pattern Analysis

To provide a richer characterization of the model's failure modes under quantization, this appendix presents a qualitative analysis of the types of reasoning errors observed across evaluation conditions. Because the automated scoring script logs only binary correct/incorrect outcomes, this analysis is based on a manual review of 20 randomly selected incorrect responses per language at each quantization level.

English Error Patterns

At FP16, English errors were predominantly attributable to genuine reasoning difficulty rather than language processing failure. The model consistently generated a valid digit (1–4) as output in all reviewed cases, indicating reliable task format comprehension. Errors clustered around questions requiring multi-step inference, negation parsing ("what would NOT be considered..."), and disambiguation between closely related answer options where two choices were both superficially plausible.

At Q8, the character of English errors shifted noticeably. A new pattern emerged that I describe here as "passage bypass". The model selected answer options that were plausible based on general world knowledge without carefully reading the specific passage. For example, on a question about accordion technique where the correct answer required extracting a specific detail from the passage, the model selected an answer consistent with general instrument-playing knowledge rather than the passage's specific claim. This pattern appeared in approximately 30% of Q8 English errors and suggests that 8-bit quantization degrades the model's ability to faithfully encode passage-specific details while preserving general reasoning ability.

At Q4, English errors concentrated heavily around complex sentence structures, embedded clauses, and logical negations. Simple declarative passages continued to be handled reasonably well. Approximately 8% of Q4 English responses produced non-numeric outputs, fragments of explanatory text that the scorer treated as incorrect, suggesting that Q4 quantization begins to degrade the model's instruction-following compliance alongside its reasoning accuracy.

Polish Error Patterns

At FP16, Polish errors mirrored the English pattern: concentrated on genuinely difficult inference questions, with consistent numeric output format. The model handled Polish

morphological complexity, including case-inflected noun phrases and verb aspect distinctions, without observable difficulty. This confirms that at full precision, cross-lingual transfer circuits are sufficiently intact to support Polish grammatical processing at a level competitive with English.

The Q8 Polish error pattern diverged from the English pattern in a way that is directly relevant to the theoretical framework. While English exhibited the passage-bypass phenomenon at Q8, Polish at Q8 showed fewer such errors despite having lower absolute accuracy. This is consistent with the hypothesis that LLM.int8()'s outlier weight preservation preferentially protects the morphological attention circuits required for Polish processing.

The Q4 Polish collapse was associated with a qualitatively distinct error type not observed at Q8 or in the English data at any level. A substantial portion of incorrect Q4 Polish responses involved apparent morphological confusion: the model selected answer options containing inflected forms of the same root word as the correct answer, but in a different grammatical case. For instance, in a question where the correct answer contained a noun in the genitive case (*stolu*), the model selected an option containing the same noun in the instrumental case (*stolem*), suggesting it had lost the ability to discriminate between case-inflected variants. This morphological confusion error type was virtually absent from Q4 English data and accounts for a disproportionate share of the Polish Q4 degradation.

This error pattern directly supports the weight sparsity mechanism proposed in Section 3.4. At 4-bit precision, the quantization grid appears to lack the resolution to maintain distinct representations for the fourteen inflected forms of a single Polish noun, effectively collapsing them into an undifferentiated cluster. For English, where a single word form covers all syntactic roles, this resolution loss has no equivalent consequence.

Conclusion of Error Analysis

The qualitative error analysis adds important texture to the quantitative findings. The morphological confusion pattern observed exclusively in Q4 Polish responses is the most direct evidence connecting precision loss to the specific structural challenge of Polish morphology. It suggests that the Convergence Effect is not simply a uniform capability reduction but a targeted dismantling of morphological disambiguation capacity, a capability that Polish relies on for basic comprehension and English does not require at all.